

TOI-2046b

R-2352

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_250727 · created 2026-08-19T01:23:27+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2460883.95 +/- 0.15 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.045
Transit depth (Rp/Rs)^2	2.4 +/- 1.4 [%]
Orbital Inclination (inc)	79.76 +/- 2.99
Airmass coefficient 1 (a1)	1.0585 +/- 0.0095
Airmass coefficient 2 (a2)	-0.037 +/- 0.086
Scatter in the residuals of the lightcurve fit is	0.9 %
Best Comparison Star	#3 - [253, 111]
Optimal Aperture	3.78
Optimal Annulus	8.76
Transit Duration (day)	0.02 +/- 0.03

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.045 vs 0.1213 (deviation 0.75σ)
Duration fit vs published	0.02 d vs 0.1033 d (deviation 2.78σ)
Mid-time O–C	-23.0 min
Depth SNR	3.4
Residual scatter	0.9 %

Light curve

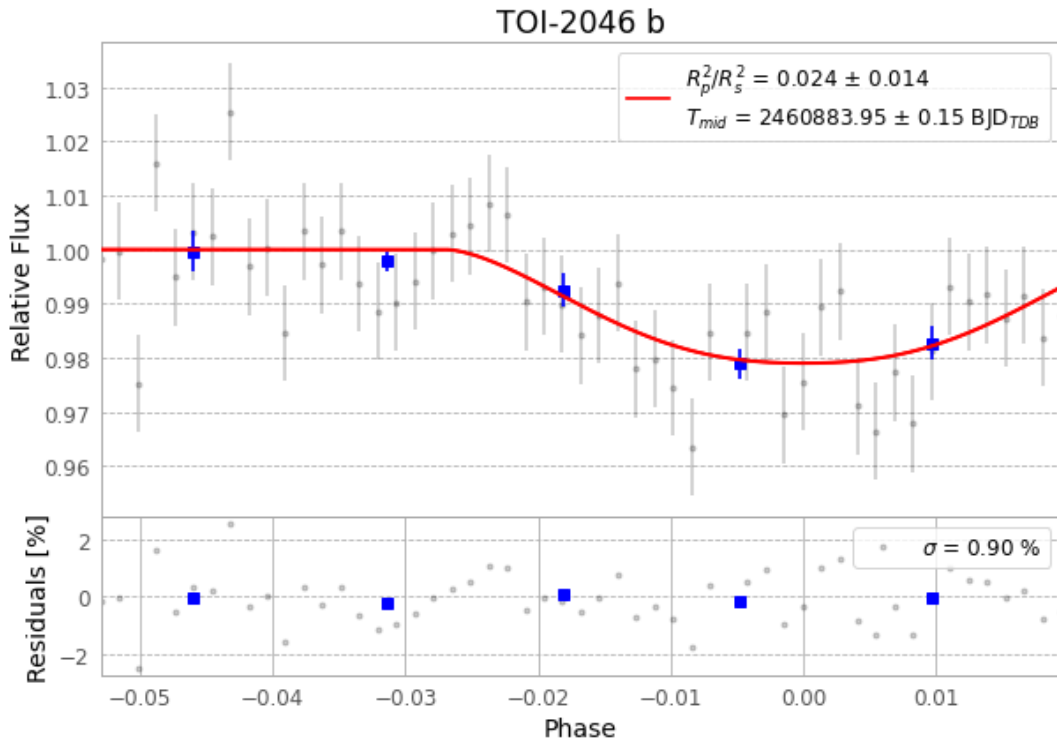


Figure 1 — FinalLightCurve_TOI-2046 b_2025-07-27.png (SHA-256 02104c01823a5a29...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [405, 219], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b950fe8eac510e4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

53 FITS frames (35 MB), TOI2046250727090016.FITS → TOI2046250727113615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-250727.log (f91fa92a28b1...), FinalLightCurve_TOI-2046 b_2025-07-27.png (02104c01823a...), FinalLightCurve_TOI-2046 b_2025-07-27.csv (76c9beb5a739...)

Compute resources

Wall time 164.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 01:23Z.